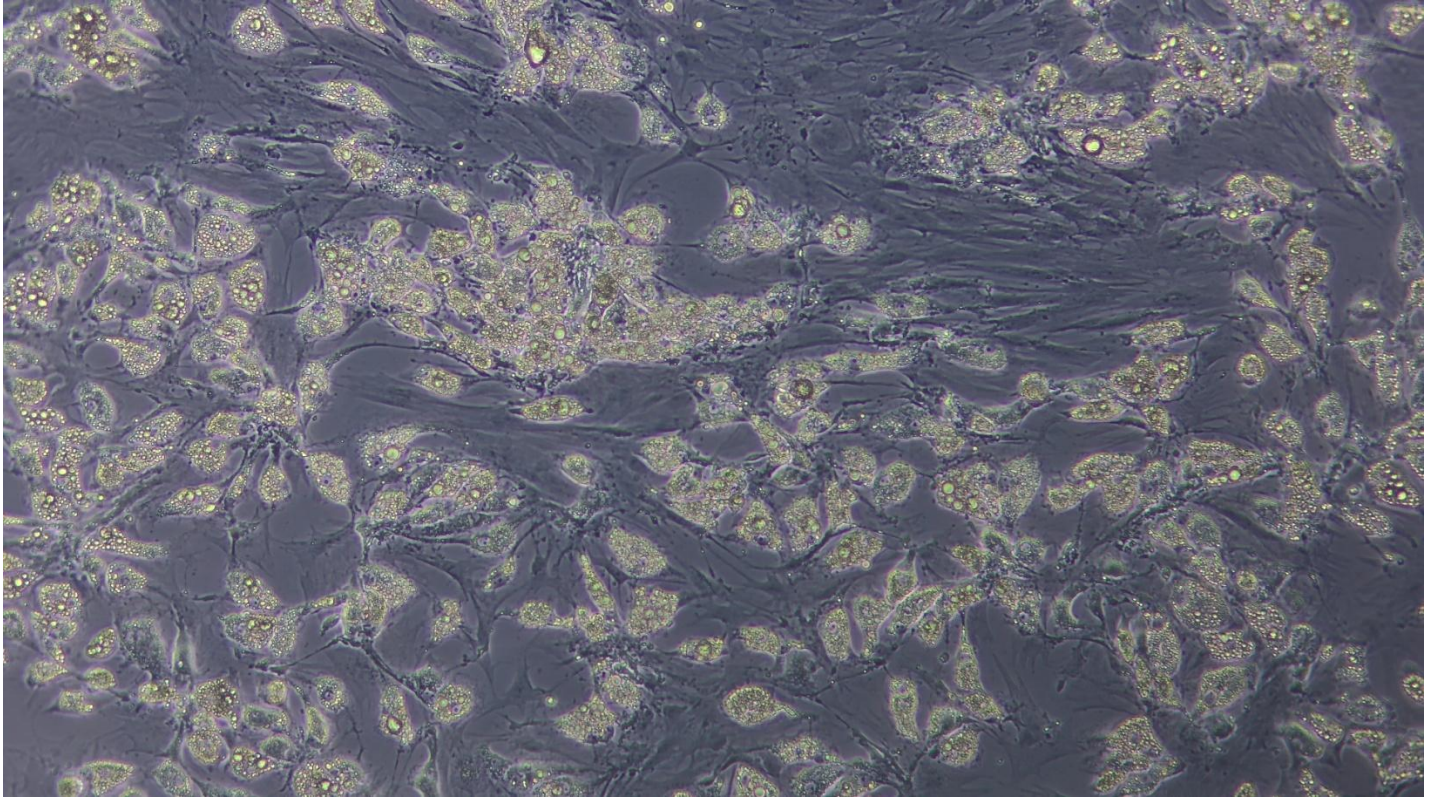


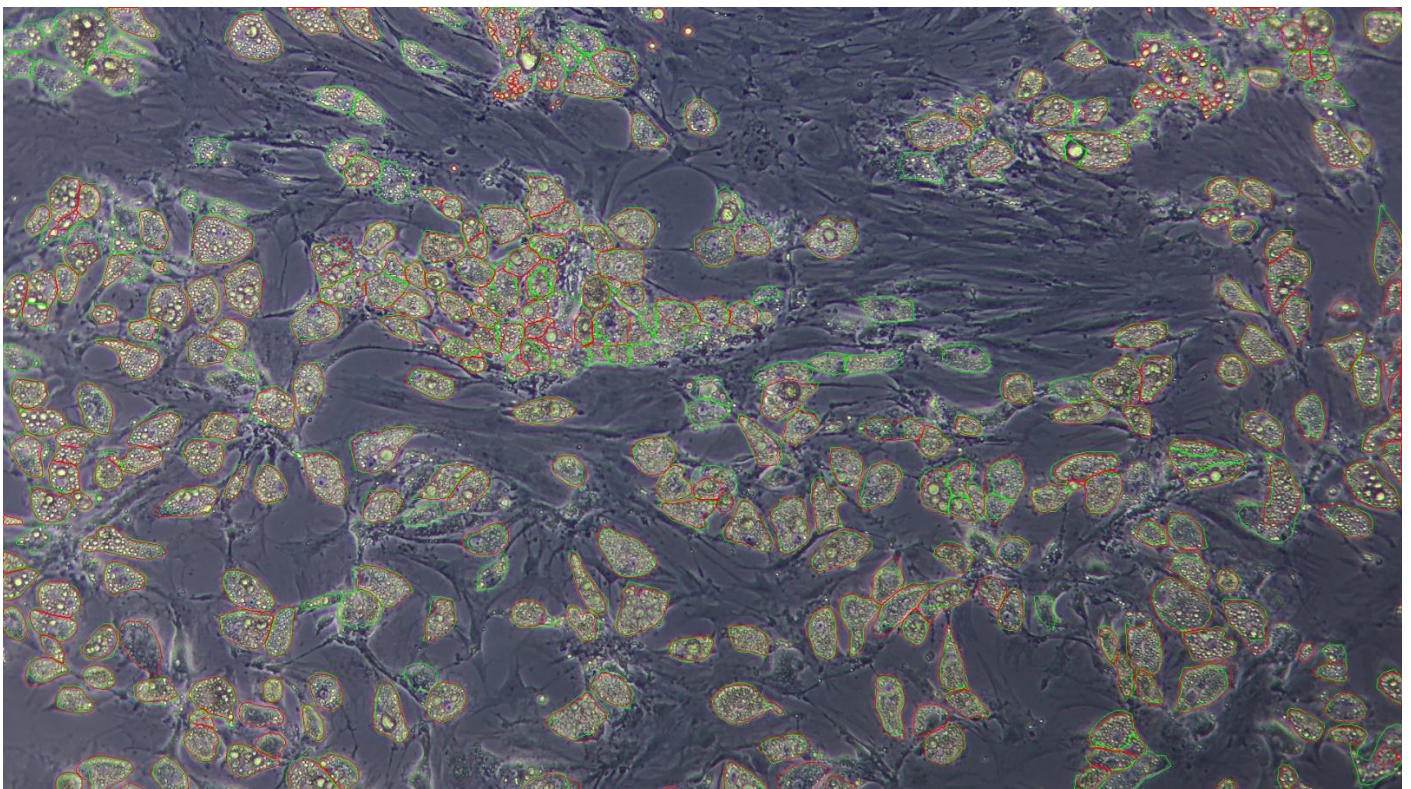
Results on one image

Positiv_16: 4 models, zero-shot, same settings (295 annotated cells)

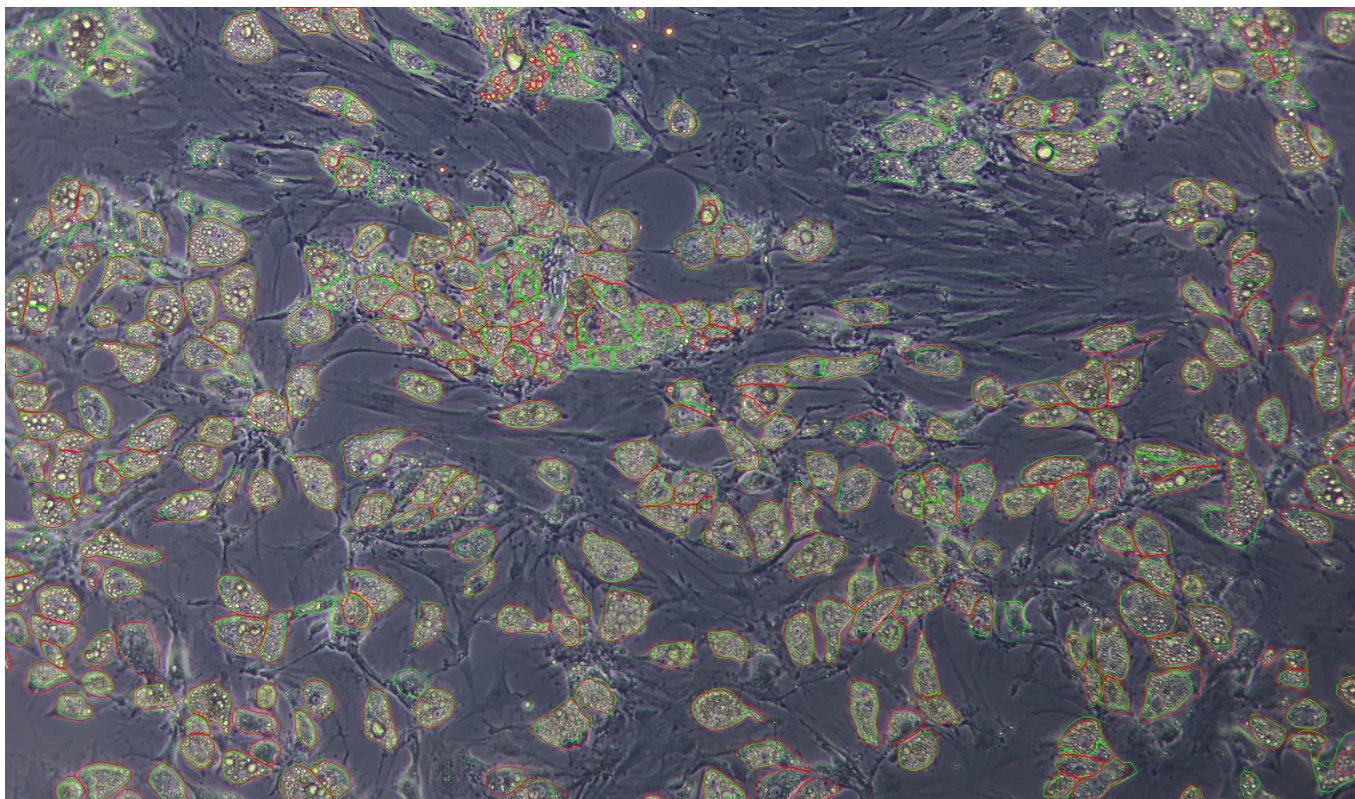
Original



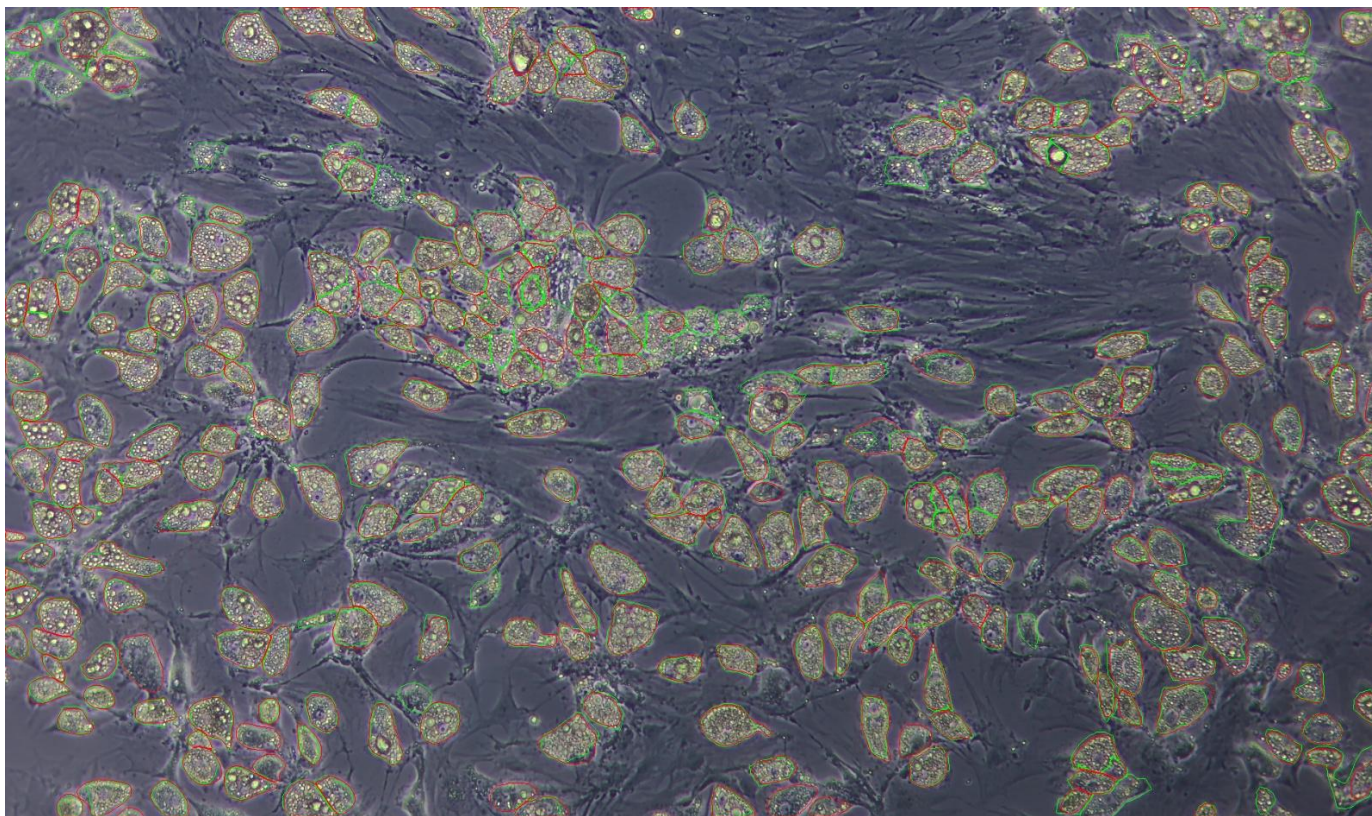
cpsam_v2 (default)



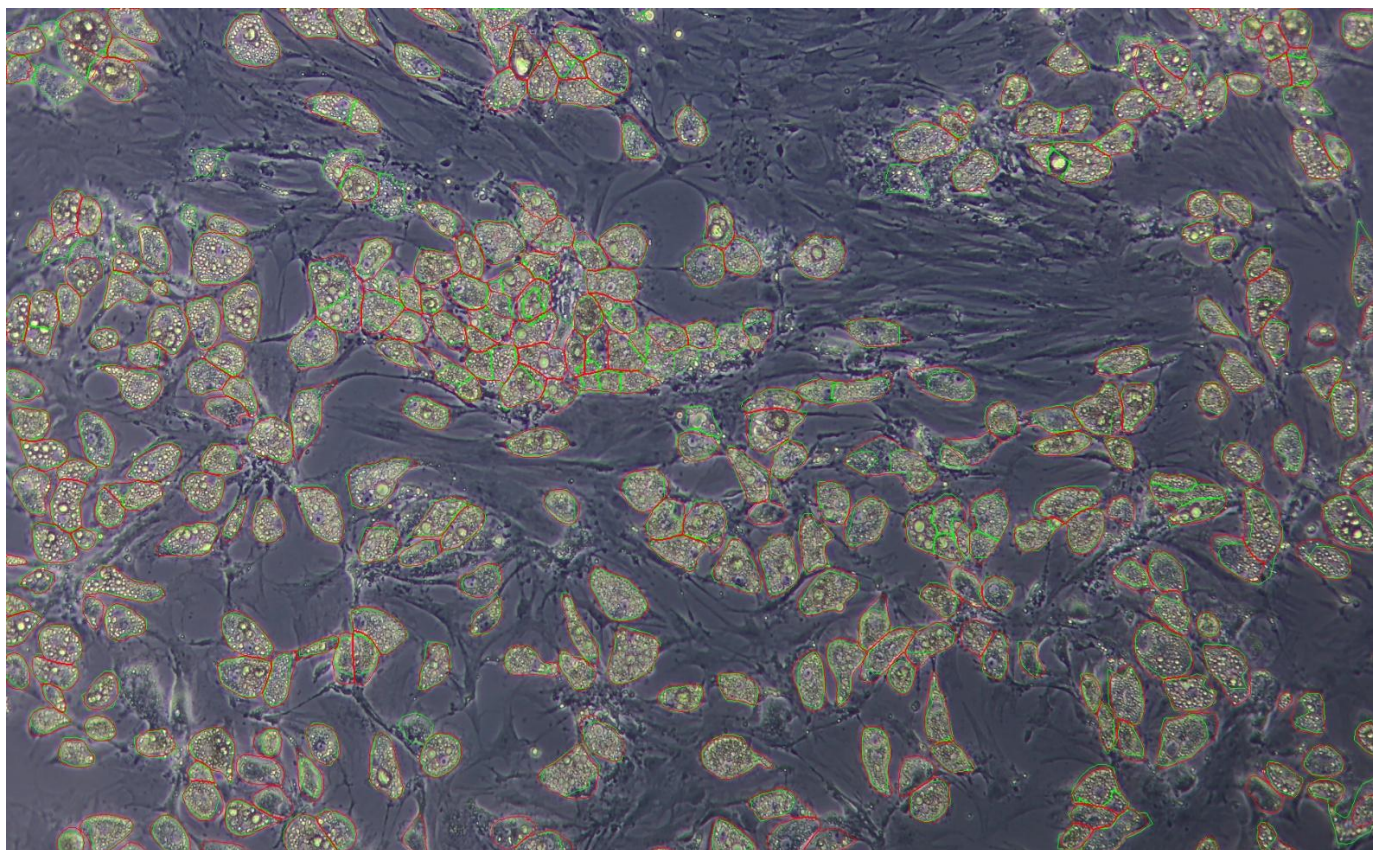
cpsam (original)



cpdino (DINOv3 ViT-L)



cpdino-vitb (DINOv3 ViT-B)



	cpsam_v2	cpsam	cpdino	cpdino-vitb
Cells found	339	320	284	304
Count error	+14.9%	+8.5%	-3.7%	+3.1%
TP / FP / FN @0.5	205 / 134 / 90	210 / 110 / 85	229 / 55 / 66	235 / 69 / 60
Precision (TP ÷ found)	0.61	0.66	0.81	0.77
Recall (TP ÷ annotated)	0.69	0.71	0.78	0.80
F1@0.5	0.647	0.683	0.791	0.785
SA@0.5	0.478	0.519	0.654	0.646
F1@0.75 (outline quality)	0.517	0.494	0.608	0.538
F1@0.9	0.054	0.033	0.086	0.033
Time / GPU memory	7.3 s / 1.60 GB	6.9 s / 1.60 GB	6.2 s / 1.29 GB	3.3 s / 0.71 GB

Term	Full form	One-line definition
Cells found (n_pred)	number of predicted cells	How many cells the model drew in the image.
n_true	number of ground-truth cells	How many cells IMES annotated by hand (295 in Positiv_16).
Count error	relative count error	How far the model's cell count is from the annotated count, in percent.
IoU	Intersection over Union	Overlap between a predicted and an annotated cell: shared area ÷ combined area (1 = identical).
TP	True Positive	A predicted cell that matches an annotated cell (correct detection).
FP	False Positive	A predicted cell that matches no annotated cell (false alarm).
FN	False Negative	An annotated cell that the model didn't find (missed cell).
@0.5 / @0.75 / @0.9	IoU threshold	A prediction counts as a match only if IoU $\geq$ 0.5 / 0.75 / 0.9.
Precision	positive predictive value	Share of the model's cells that are real: $TP \div \text{found}$.
Recall	sensitivity / true positive rate	Share of the real cells that the model found: $TP \div \text{annotated}$.
F1	F1 score (harmonic mean of precision and recall)	One score balancing false alarms and misses: $2TP \div (2TP + FP + FN)$.
SA	Segmentation Accuracy (Cellpose calls it "AP")	Stricter combined score: $TP \div (TP + FP + FN)$. Always equals $F1 \div (2 - F1)$.
F1@0.75	F1 at IoU threshold 0.75	F1 with a stricter match rule, so it measures outline accuracy.
F1@0.9	F1 at IoU threshold 0.9	F1 with a very strict rule. Mostly limited by how precise the hand annotation is.
Time	inference time	Seconds the model needed to segment the image (measured, not calculated).
GPU memory	peak GPU memory	Most graphics-card memory the model used (measured, not calculated).

All examples use n_true = 295 (annotated cells in Positiv_16).

The three metric levels in these files:

- **@0.5:** a predicted cell counts as correct if it overlaps the annotated cell by at least 50% (IoU $\geq$ 0.5). This answers “was the cell found?”
- **@0.75:** the overlap must be at least 75%. This answers “is the outline accurate?”, which matters for measuring cell size.
- **@0.9:** the overlap must be at least 90%. All models score almost zero here (0.03–0.09). That’s normal: hand-drawn outlines aren’t precise to that level, so this threshold tests the annotator as much as the model. **Ignore it.**
- **SA** is a stricter version of F1 ($SA = F1 \div (2 - F1)$). It always ranks the models in the same order, so it adds no new information.

Calculations, row by row

Cells found (n_pred)

Read directly from the output, not calculated: **339 / 320 / 284 / 304**.

Check: TP + FP must equal cells found.

Model	TP + FP
cpsam_v2	205 + 134 = 339 ✓
cpsam	210 + 110 = 320 ✓
cpdino	229 + 55 = 284 ✓
cpdino-vitb	235 + 69 = 304 ✓

And TP + FN must equal 295:

Model	TP + FN
cpsam_v2	205 + 90 = 295 ✓
cpsam	210 + 85 = 295 ✓
cpdino	229 + 66 = 295 ✓
cpdino-vitb	235 + 60 = 295 ✓

Count error = (found – annotated) ÷ annotated × 100%

Model	Calculation	Result
cpsam_v2	$(339 - 295) \div 295 = 44 \div 295$	+14.9%
cpsam	$(320 - 295) \div 295 = 25 \div 295$	+8.5%
cpdino	$(284 - 295) \div 295 = -11 \div 295$	-3.7%
cpdino-vitb	$(304 - 295) \div 295 = 9 \div 295$	+3.1%

TP / FP / FN @0.5

Counted by the matching code, not calculated. Each predicted cell is paired with at most one annotated cell, and a pair counts as TP only if IoU ≥ 0.5. Unpaired predictions are FP, unpaired annotated cells are FN.

Model	TP	FP = found – TP	FN = 295 – TP
cpsam_v2	205	$339 - 205 = 134$	$295 - 205 = 90$
cpsam	210	$320 - 210 = 110$	$295 - 210 = 85$
cpdino	229	$284 - 229 = 55$	$295 - 229 = 66$
cpdino-vitb	235	$304 - 235 = 69$	$295 - 235 = 60$

Precision = TP ÷ cells found

Model	Calculation	Result
cpsam_v2	$205 \div 339$	$0.605 \rightarrow \mathbf{0.61}$
cpsam	$210 \div 320$	$0.656 \rightarrow \mathbf{0.66}$
cpdino	$229 \div 284$	$0.806 \rightarrow \mathbf{0.81}$
cpdino-vitb	$235 \div 304$	$0.773 \rightarrow \mathbf{0.77}$

Recall = TP ÷ annotated (295)

Model	Calculation	Result
cpsam_v2	$205 \div 295$	$0.695 \rightarrow \mathbf{0.69}$
cpsam	$210 \div 295$	$0.712 \rightarrow \mathbf{0.71}$
cpdino	$229 \div 295$	$0.776 \rightarrow \mathbf{0.78}$

Model	Calculation	Result
cpdino-vitb	$235 \div 295$	$0.797 \rightarrow \mathbf{0.80}$

$$\mathbf{F1@0.5 = 2 \cdot TP \div (2 \cdot TP + FP + FN)}$$

Model	Calculation	Result
cpsam_v2	$2 \cdot 205 \div (410 + 134 + 90) = 410 \div 634$	0.647
cpsam	$2 \cdot 210 \div (420 + 110 + 85) = 420 \div 615$	0.683
cpdino	$2 \cdot 229 \div (458 + 55 + 66) = 458 \div 579$	0.791
cpdino-vitb	$2 \cdot 235 \div (470 + 69 + 60) = 470 \div 599$	0.785

Check via precision and recall: $F1 = 2 \cdot P \cdot R \div (P + R)$.

Model	Calculation	Result
cpsam_v2	$2 \times 0.605 \times 0.695 \div 1.300$	0.647 ✓
cpsam	$2 \times 0.656 \times 0.712 \div 1.368$	0.683 ✓
cpdino	$2 \times 0.806 \times 0.776 \div 1.582$	0.791 ✓
cpdino-vitb	$2 \times 0.773 \times 0.797 \div 1.570$	0.785 ✓

Shortcut: the denominator $2TP + FP + FN$ always equals found + annotated. For cpdino: $284 + 295 = 579$.

$$\mathbf{SA@0.5 = TP \div (TP + FP + FN)}$$

Model	Calculation	Result	Check: $F1 \div (2 - F1)$
cpsam_v2	$205 \div (205 + 134 + 90) = 205 \div 429$	0.478	$0.647 \div 1.353 = 0.478 \checkmark$
cpsam	$210 \div (210 + 110 + 85) = 210 \div 405$	0.519	$0.683 \div 1.317 = 0.519 \checkmark$
cpdino	$229 \div (229 + 55 + 66) = 229 \div 350$	0.654	$0.791 \div 1.209 = 0.654 \checkmark$
cpdino-vitb	$235 \div (235 + 69 + 60) = 235 \div 364$	0.646	$0.785 \div 1.215 = 0.646 \checkmark$

F1@0.75 = the same formula with TP / FP / FN counted at IoU $\geq$ 0.75

Model	TP / FP / FN @0.75	Calculation	Result
cpsam_v2	164 / 175 / 131	$2 \cdot 164 \div (328 + 175 + 131) = 328 \div 634$	0.517
cpsam	152 / 168 / 143	$2 \cdot 152 \div (304 + 168 + 143) = 304 \div 615$	0.494
cpdino	176 / 108 / 119	$2 \cdot 176 \div (352 + 108 + 119) = 352 \div 579$	0.608
cpdino-vitb	161 / 143 / 134	$2 \cdot 161 \div (322 + 143 + 134) = 322 \div 599$	0.538

The denominators (634, 615, 579, 599) are the same as at @0.5, because they always equal found + annotated. Only TP changes with the threshold.

F1@0.9 = the same formula at IoU $\geq$ 0.9

Model	TP / FP / FN @0.9	Calculation	Result
cpsam_v2	17 / 322 / 278	$2 \cdot 17 \div 634 = 34 \div 634$	0.054
cpsam	10 / 310 / 285	$2 \cdot 10 \div 615 = 20 \div 615$	0.033
cpdino	25 / 259 / 270	$2 \cdot 25 \div 579 = 50 \div 579$	0.086
cpdino-vitb	10 / 294 / 285	$2 \cdot 10 \div 599 = 20 \div 599$	0.033

Time / GPU memory

Measured by the script while the model ran; nothing is calculated.

Model	Time	GPU memory
cpsam_v2	7.3 s	1.60 GB
cpsam	6.9 s	1.60 GB
cpdino	6.2 s	1.29 GB
cpdino-vitb	3.3 s	0.71 GB

Times from single runs vary between runs (the later batch run measured about 3.6 s and 1.7 s).

Compare speeds only within one run.

Table note: in the precision row, cpdino's **0.81** (bold) is the best value. In the F1@0.5 row, cpdino (0.791) and cpdino-vitb (0.785) differ by only 0.006, which is within the noise of a single image (about ± 0.025). Treat it as a tie.

Analysis

1. **The DINO models are clearly better on this image:** F1 0.79 vs 0.65–0.68. The difference is about 0.10–0.14. On one image with about 300 cells, the noise is about ± 0.025 , so this gap is real for this image.
2. **The gain comes mainly from fewer false detections.** cpsam_v2 has 134 false objects, cpdino only 55. The DINO models also miss fewer cells (60–66 vs 85–90).
3. **cpdino vs cpdino-vitb:**
 - **Finding cells (F1@0.5):** a tie (0.791 vs 0.785, a difference of 0.006, well within the noise).
 - **Outlines (F1@0.75):** cpdino is clearly better (0.608 vs 0.538). It matches the annotated cell shape more closely.
 - **Error pattern:** cpdino has fewer false detections (55 vs 69); cpdino-vitb misses fewer cells (60 vs 66).
 - **Count:** both are close to the truth (–3.7% and +3.1%).
4. **cpsam vs cpsam_v2:** the original weights are better here (F1 0.683 vs 0.647, fewer false detections). Newer isn't better for these cells.
5. **Efficiency:** cpdino-vitb is about 2× faster than the others and needs less than half the GPU memory.

Verdict for this image

1. **cpdino: best overall.** It ties for best at finding cells, draws clearly the most accurate outlines, and has the fewest false detections.
2. **cpdino-vitb: close second.** Same detection quality and the most accurate cell count, about 2× faster and lighter, but less precise outlines.
3. **cpsam:** clearly behind; it over-segments.
4. **cpsam_v2:** weakest; the most false detections and the worst count (+15%).